MELCHIOR JORDA

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PROFESSIONAL PROFILE

Full-stack developer building mobile and web applications in TypeScript, with strong systems foundations in C/C++ from Ecole 42. Creator of **NEXUS**, a self-hosted AI personal assistant for Android (React Native + Fastify), now in alpha. Based in Malaga, open to opportunities in France and Spain.

FEATURED PROJECT

NEXUS - Self-Hosted AI Personal Assistant (Android)

June 2026 - Present (Alpha) | Personal project

- Designed and built, with AI-assisted development (**Claude Code**), an offline-first Android app (**React Native/Expo**, **TypeScript**) and its API (**Node.js/Fastify**): meal logging by photo and voice, weight and energy tracking, strength-training coach, notes and knowledge graph.
- Offline-first sync: local WatermelonDB database with a sync queue to a self-hosted server (**PostgreSQL**/Prisma, **Neo4j**, Redis, Docker Compose).
- AI features: meal recognition from photos, voice dictation (self-hosted Whisper), coaching through the **Gemini** and **Claude** APIs, with safety guardrails on health topics.
- Health Connect integration (smartwatch workouts, steps, smart scale) feeding an energy-expenditure model.
- Product-ready: FR/EN/ES localisation, invitation-only signup, GDPR account deletion, in-app feedback, update channel. 400+ commits, ~120k lines of TypeScript, 90 test files.

TECHNICAL SKILLS

Programming languages : C, C++ (STL, templates, OOP), JavaScript (ES6+, async/await), TypeScript, PHP, Java

Web technologies : Vue.js, Angular, Symfony (Doctrine, Twig, Security), Node.js, HTML5, CSS3, Bootstrap, RESTful API

Databases : Firestore (NoSQL), MySQL

AI : LLM API integration (Gemini, Claude), prompt engineering and structured JSON outputs, AI-assisted development (Claude Code)

Systems programming : POSIX threads, mutex synchronisation, process management, Unix system calls, memory management

Development tools : Git, GitHub, Valgrind, GDB, Apache, Firebase, Bitbucket, Trello, Figma, Gradle (Android builds), Tailscale

Software engineering : Object-oriented design, data structures, algorithms, code review, agile methodologies, internationalisation (i18n)

Languages : French (Native) | English (Fluent) | Spanish (Intermediate)

PROFESSIONAL EXPERIENCE

Linguistic Quality Assurance (LQA) Tester | Localsoft, Malaga, Spain

January 2025 - Present

- LQA testing on 15+ localisation projects: translation errors, cultural inconsistencies and display issues.
- Collaboration with international teams to ensure quality standards and deliver comprehensive test reports.
- Team coordination (5-6 members) during managerial absences, organising workflows and maintaining schedules.

Web Developer Intern | Ecole ENI Informatique, France

January 2024 - March 2024

- Development of a quiz application and an event management website (**Node.js, Electron.js, jQuery, MySQL**).
- Autonomous end-to-end project management, collaboration via **Bitbucket** and **Trello** using **agile methodology.**

EDUCATION

Ecole 42 Malaga | Software Engineering Programme - Common Core, advanced stage (In progress)

September 2024 - Present | Malaga, Spain

- Intensive peer-to-peer curriculum: systems C/C++, algorithms, Unix, 3D graphics, concurrent and network programming.

Ecole ENI Informatique | Web & Mobile Development Diploma

2023 - 2024 | France

- PHP/Symfony, Angular, Java, MySQL, HTML/CSS. Projects: event management and full-stack auction platform.

HACKATHONS & COMPETITIONS

1st Prize - NASA Space Apps Challenge Malaga 2025

October 2025 | https://re-fresh.earth

- 1st place with 'Re-Fresh Earth': intelligent urban planning tool using NASA satellite data to monitor natural resources and infrastructure.
- Role: frontend development of the website, in a team of 4 Ecole 42 students.
- Technologies: JavaScript, web APIs, data visualisation, geospatial mapping

PERSONAL PROJECTS

NainVert - Psychedelic Streetwear E-Commerce Platform

February 2026 - Present (In progress) | Personal project

- **Vue.js** + **Firebase** (Firestore, Cloud Functions): real-time inventory, secure payments, role-based admin dashboard.

KEY TECHNICAL PROJECTS (Ecole 42)

Minishell - Unix Shell Implementation

June - October 2025 | github.com/M-Jorda/Minishell

- Functional Unix shell in C: command parsing, pipes, redirections, process management (fork/exec) and signals (SIGINT, SIGQUIT).

Philosophers - Concurrent Programming Simulation

October - November 2025 | github.com/M-Jorda/philosophers

- Dining philosophers problem with POSIX threads, mutexes and semaphores: deadlock prevention and race condition management.

cub3D - 3D Ray-Casting Graphics Engine

November 2025 (2 weeks)

- 3D rendering engine in C (Wolfenstein-style): texture mapping, collision detection, mathematical transformations (~4,000 lines).

C++ OOP Modules (CPP00-04)

December 2025 - January 2026 | github.com/M-Jorda/CPP

- Solid grounding in C++ fundamentals: inheritance, polymorphism, templates, STL, exceptions and design patterns.

C++ Advanced Modules (CPP05-09)

June 2026 - September 2026 | github.com/M-Jorda/CPP

- Advanced C++98: exceptions, type casting, serialization, templates, STL containers, iterators and algorithms.

ADDITIONAL INFORMATION

Availability: ASAP
Location: Based in Malaga, open to France/Spain
Portfolio: melchior-jorda.online | GitHub repositories with source code and documentation